

Module 8: Cellular Genetics — Keys to Success

Learning Objectives

By the end of this module, you should be able to:

1. Describe the stages of the cell cycle (interphase, mitosis, cytokinesis)
2. Sequence and describe the phases of mitosis (PMAT)
3. Sequence and describe the phases of meiosis I and meiosis II
4. Compare and contrast mitosis and meiosis (purpose, products, ploidy)
5. Explain the three sources of genetic variation in meiosis
6. Describe chromosome structure: chromatin, chromosomes, sister chromatids, homologs
7. Explain cell cycle checkpoints and how their failure leads to cancer
8. Explain nondisjunction and its consequences (aneuploidy)

Key Terms to Know

Chromosome Organization

- **Chromatin** — Loosely packed DNA + histone proteins (interphase form)
- **Chromosome** — Tightly condensed chromatin (visible during division)
- **Sister Chromatids** — Two identical copies of one chromosome, joined at the centromere (produced during S phase)
- **Centromere** — Region where sister chromatids are attached
- **Homologous Chromosomes (Homologs)** — Matching chromosome pairs (one from each parent); same genes, possibly different alleles
- **Diploid (2n)** — Having two complete sets of chromosomes (somatic cells); humans: $2n = 46$
- **Haploid (n)** — Having one set of chromosomes (gametes); humans: $n = 23$
- **Autosome** — Any chromosome that is not a sex chromosome (humans: 22 pairs)
- **Sex Chromosome** — X and Y chromosomes that determine biological sex

Cell Cycle and Mitosis

- **Interphase** — G₁ (growth) → S (DNA synthesis/replication) → G₂ (preparation for division)
- **Mitosis** — Nuclear division: Prophase → Metaphase → Anaphase → Telophase (PMAT)
- **Cytokinesis** — Division of the cytoplasm; animal cells: cleavage furrow; plant cells: cell plate
- **Spindle Fibers** — Microtubules that move chromosomes during division
- **Centrioles** — Organize spindle fibers in animal cells
- **Cell Plate** — Structure that divides plant cells during cytokinesis
- **Cleavage Furrow** — Pinching of cell membrane that divides animal cells

Meiosis

- **Meiosis** — Two-division process producing four haploid cells from one diploid cell
- **Synapsis** — Pairing of homologous chromosomes during Prophase I
- **Tetrad (Bivalent)** — The four-chromatid structure formed by synapsed homologs
- **Crossing Over** — Exchange of DNA segments between non-sister chromatids of homologs during Prophase I
- **Independent Assortment** — Random orientation of homologous pairs at Metaphase I
- **Random Fertilization** — Any sperm can fertilize any egg → enormous combinatorial diversity
- **Nondisjunction** — Failure of chromosomes to separate properly → aneuploidy
- **Aneuploidy** — Abnormal chromosome number (e.g., trisomy 21 = Down syndrome)

Cancer

- **Tumor Suppressor Gene** — Gene that inhibits cell division (e.g., p53, Rb); loss → uncontrolled growth
- **Proto-oncogene** — Normal gene promoting cell growth; mutation → **oncogene** (constitutively active)
- **Metastasis** — Spread of cancer cells to other body sites

Mitosis: Phase by Phase

Phase	What Happens
Prophase	Chromatin condenses into chromosomes; nuclear envelope begins to break down; spindle forms
Metaphase	Chromosomes line up at the cell's equator (metaphase plate)
Anaphase	Sister chromatids separate and move to opposite poles
Telophase	Nuclear envelopes reform; chromosomes decondense; spindle disassembles
Cytokinesis	Cytoplasm divides → two identical diploid daughter cells

Meiosis: Phase by Phase

Meiosis I — Reductional Division ($2n \rightarrow n$)

Phase	What Happens
Prophase I	Homologs pair (synapsis) forming tetrads; crossing over occurs; nuclear envelope breaks down
Metaphase I	Homologous pairs line up at equator (independent assortment — random orientation)
Anaphase I	Homologs separate (not sister chromatids); one homolog goes to each pole
Telophase I	Two haploid cells form; each has one chromosome from each homologous pair

Meiosis II — Equational Division ($n \rightarrow n$)

Phase	What Happens
Prophase II	Chromosomes condense; spindle reforms
Metaphase II	Chromosomes line up at equator
Anaphase II	Sister chromatids separate
Telophase II	Four haploid daughter cells form — each is genetically unique

Mitosis vs. Meiosis: Key Comparison

Feature	Mitosis	Meiosis
Divisions	1	2
Daughter Cells	2	4
Ploidy	Diploid ($2n$)	Haploid (n)
Genetic Result	Identical to parent	Genetically unique
Purpose	Growth, repair, asexual reproduction	Produce gametes (sex cells)
Crossing Over	No	Yes (Prophase I)
Independent Assortment	No	Yes (Metaphase I)

Three Sources of Genetic Variation in Meiosis

1. **Crossing over** (Prophase I) — Swaps DNA between homologs → new allele combinations

2. **Independent assortment** (Metaphase I) — Random orientation of each homologous pair → $2^{23} = \sim 8.4$ million combinations per gamete in humans
3. **Random fertilization** — Any sperm $\times$ any egg → $8.4 \text{ million} \times 8.4 \text{ million} > 70$ trillion unique combinations

Cell Cycle Checkpoints and Cancer

- **G₁ Checkpoint** — Is the cell large enough? Is DNA undamaged? Nutrients available?
- **G₂ Checkpoint** — Is DNA replication complete? Any DNA damage?
- **M Checkpoint (Spindle Checkpoint)** — Are all chromosomes properly attached to spindle fibers?

When checkpoints fail → uncontrolled cell division → **cancer**

Study Tips

1. **PMAT** — Prophase, Metaphase, Anaphase, Telophase (works for both mitosis and meiosis)
2. **"Homologs in I, Sisters in II"** — Meiosis I separates homologs; Meiosis II separates sister chromatids
3. **Draw side-by-side diagrams** — Show a cell going through mitosis alongside a cell going through meiosis
4. **Think about ploidy** — Start at $2n$, track what happens at each stage
5. **Connect to genetics** — Crossing over and independent assortment are WHY siblings look different
6. **Cancer = checkpoint failure** — Know p53, Rb, and the concepts of proto-oncogene → oncogene